Jethro Phuah An Ping

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PROFILE

Senior Software Engineer with 3+ years of experience building and operating production LLM and multi-agent systems at DSTA. Hands-on across Python, FastAPI, LangChain, LangGraph, Deep Agents, MCP, RAG, context engineering and evaluation, with experience translating ambiguous user workflows into agent architectures and taking them from prototype through production. Comfortable working directly with users and engineering teams to prototype solutions, make architectural tradeoffs and turn real operational problems into reliable production systems.

PROFESSIONAL EXPERIENCE

Defence Science and Technology Agency (DSTA) — SG Government Agency for Defence *Jul 2023 – Present*

Senior Software Engineer | Lead Engineer, CTI Platform

- Architected and shipped a production multi-agent CTI Research Agent using LangChain, LangGraph and Deep Agents, designing agent state, task routing, context management, tool selection and multi-step execution across research, retrieval and internal data workflows.
- Evolved the agent architecture from fixed workflows into reusable skills and dynamically composed agents, allowing capabilities and tools to be selected based on the task as use cases expanded across threat research and detection engineering.
- Developed MCP servers, custom agent tools and reusable Skills.md capabilities connecting agents to enterprise APIs, internal databases, RAG search, enrichment services and platform actions through structured interfaces.
- Established a 50-question evaluation framework measuring accuracy, safety and latency, with retrieval validation, prompt injection testing, output guardrails and an independent validator agent; used production queries and analyst feedback to drive iteration, reducing hallucinations by ~40%.
- Engineered RAG across 400K+ records and 15GB+ of enterprise knowledge, combining dense retrieval, Elasticsearch/BM25, reranking, metadata filtering and contextual chunk expansion to handle both semantic research and exact identifiers.
- Implemented persistent agent workflows with PostgreSQL-backed sessions, conversation history and research outputs, maintaining user context across requests and supporting continued research over multiple interactions.
- Partnered directly with analysts to translate ambiguous research workflows into technical solutions, rapidly prototyping with users and deciding where to use agentic AI, deterministic automation or human review based on reliability and operational requirements.
- Automated an analyst workflow with LLM-powered alert generation, IOC extraction and enrichment across ~7 external sources, reducing processing time from ~2 hours to ~5 minutes.
- Deploy and operate 10+ Docker services on AWS, owning CI/CD, testing, logging, monitoring, API gateway configuration and production troubleshooting across agent, model, retrieval and data services.

EDUCATION

Nanyang Technological University *Aug 2019 – May 2023*

Double Degree in Business and Computer Science — Specialisation in Business Analytics and Machine Learning
GPA: 4.63 / 5.00 (Computer Science) — First Class Honours | 4.49 / 5.00 (Business) — Second Class Honours (Upper Division)

TECHNICAL SKILLS

Agent Engineering: LangChain, LangGraph, Deep Agents, MCP, Agent Skills, Multi-Agent Systems, Context Engineering, LLM Evaluation
AI & Retrieval: RAG, vLLM, Qwen, BGE, spaCy, Milvus, Elasticsearch/BM25, Neo4j
Software & Data: Python, FastAPI, TypeScript, React, Node.js, GraphQL, REST APIs, PostgreSQL, Redis, RabbitMQ

Cloud & DevOps: AWS, Docker, CI/CD, Kong API Gateway, Git/GitHub
AI Development: Claude Code, Codex, GitHub Copilot